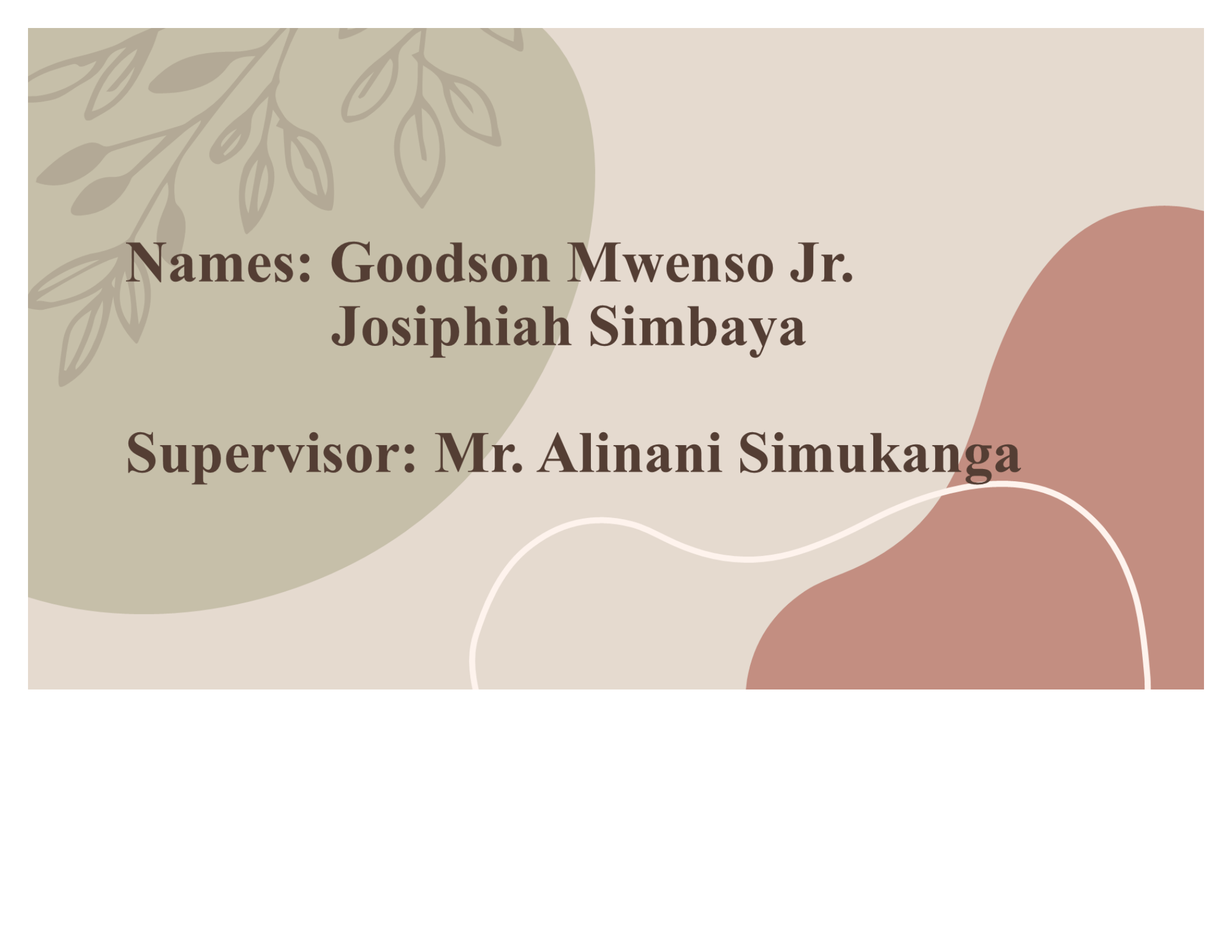




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Project Proposal: Butchery Inventory & Stock Management System

Smart Butchery Inventory Management System (SBIMS)

1. Introduction

Butcherries handle perishable products, variable weights, and frequent daily transactions. Managing stock manually often leads to inaccurate records, spoilage losses, and poor tracking of meat cuts and sales.

This project proposes a **Smart Butchery Inventory Management System** that includes:

An **Android application** for cashiers and shop attendants

A **web based system** for management oversight

The system is designed specifically to handle **weight-based inventory, perishable goods, and real-time sales tracking**.

2. Problem Statement

Traditional butchery operations face challenges such as:

Difficulty tracking meat by **weight (kg) instead of units**

Losses due to **spoilage and poor stock rotation**

Lack of visibility on **daily sales and stock movement**

Errors in manual recording of stock in/out

No proper tracking of **different meat cuts (e.g., ribs, steak, mince)**

This results in **financial losses, inefficiency, and lack of accountability.**

3. Objectives

Main Objective

To develop a system that enables efficient, accurate, and real-time management of butchery inventory.

Specific Objectives

Track stock based on **weight (kg)** instead of just quantity

Enable cashiers to **record sales and deduct stock instantly**

Allow management to monitor **stock levels and sales remotely**

Reduce losses due to **spoilage and mismanagement**

Provide **reports on sales, stock usage, and wastage**

4. System Overview

A. Android Application (Cashiers / Shop Attendants)

Features:

- Secure login system
- Select meat type (e.g., beef, chicken, pork)
- Input weight sold (e.g., 2.5 kg)
- Automatic stock deduction
- Add new stock (e.g., fresh deliveries)
- View available stock (simplified view)
- Quick daily sales recording

B. Web Application (Management)

Features:

- Dashboard showing:
 - Total stock by meat type
 - Daily sales
 - Low-stock alerts
- Manage product categories (cuts of meat)
- Monitor stock inflow (deliveries) and outflow (sales)
- Track **wastage/spoilage**
- Generate reports:
 - Daily/weekly sales
 - Profit estimation
 - Stock usage trends
- User management (cashiers, admin)

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5. Key System Features (Butchery-Specific)

•Weight-Based Inventory Tracking

- Stock managed in kilograms instead of item count

•Meat Cut Categorization

- Example:
 - Beef → steak, ribs, mince
 - Chicken → wings, thighs, breast

•Spoilage/Wastage Tracking

- Record expired or spoiled meat
- Helps reduce losses and improve planning

•Barcode/QR Code Scanning

- Scan packaged meat for quick identification
- Automatically retrieve product details (cut type, price per kg, expiry date)
- Reduce manual input errors and speed up checkout.

•Low Stock Alerts

- Notify when stock drops below a threshold

•Daily Sales Summary

- Total kg sold
 - Revenue generated
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Existing Systems

- Many butcheries use manual record keeping (notebooks or spreadsheets)
- Some use general POS (Point of Sale) systems designed for supermarkets
- Retail systems often use barcode scanners for packaged goods

Technologies in Use

- Barcode/QR code scanning in retail environments
- Digital weighing scales for measuring products
- Mobile and web-based inventory systems

Limitations of Existing Systems

- Do not support weight-based inventory tracking effectively
- Lack spoilage/expiry tracking for perishable goods
- Not designed for different meat cuts and categories
- No integration with weighing scale devices
- Often too complex or not tailored for small butcheries

Identified Gap

- No specialized system that combines:
 - Weight based tracking
 - Perishability management
- Real-time stock updates
- Barcode and scale integration

Proposed Solution

- A Smart Butchery Inventory Management System
 - Designed specifically for butcheries
 - Enables accurate, real-time stock management
- Integrates barcode scanning and weighing scales

6. System Architecture

- **Android App:** Java
- **Web App:** HTML, CSS, JavaScript
- **Backend:** Node.js
- **Database:** MySQL
- **API:** RESTful API for communication

7. Expected Outcomes

- Accurate tracking of meat stock by weight
- Reduced losses from spoilage and errors
- Faster transaction handling at the butchery
- Better visibility of business performance
- Improved decision-making for restocking

8. Tools & Technologies

- Android Studio
- VS Code
- MySQL
- GitHub
- Postman (API testing)

BUDGET

Android Studio, VS Code (Free) = K0

Database Hosting Firebase / MySQL Hosting = K950

Barcode Scanner Handheld scanner device = K760

Digital Weighing Scale With connectivity support = K1580

Internet & Testing connectivity =K100

Miscellaneous Printing and documentation =K300

TOTAL: K3690

9. Project Timeline (10–12 Weeks)

Phase	Duration
Requirements Analysis	1 week
System Design (UML, DB design)	2 weeks
Backend Development	3 weeks
Android Development	3 weeks
Web Dashboard Development	2 weeks
Testing & Debugging	1 week

Literature Review:

- NPOS (NETONE POINT OF SALE)
- SUPERMARKET POINT OF SALES

References:

- ZAMBEEF OUTLETS
- MEATCON
- SHOPRITE BUTCHERY STAFF

10. Future Enhancements

- Barcode/QR code for packaged meat
- Integration with weighing scale devices
- Expiry date tracking with alerts
- Multi-branch butchery management
- Mobile app for management

11. Conclusion

This system addresses the unique challenges of butchery operations by focusing on **weight-based tracking, perishability, and real-time monitoring**. It will improve efficiency, reduce losses, and provide valuable insights into daily operations.



THE END!